COMP-4150

Project Requirement Specification

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Team Members

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Functional Requirements

Use Case 1: Search for Books

Actors: Customer Patron, or staff member

Precondition: Book records are in the database with assigned titles, authors, ISBNs and categories.

Steps:

1. The user searches by using ISBN, author, or book title.

2. The system queries database and displays matching books showing price, inventory and availability and will ignore capitalizes letters and random spaces.

3. The user selects the book

4. The system will display prices and quantity.

Alternative Outcome: If the system can't find the book, it will display that no results were found and try again.

Post condition: The user can see all the information without changing anything in the database

Use Case 2: Place an Order

Actors: Customer

Precondition: Customer is logged in and selected a book aavailable to purchase

Steps:

1. The customer adds a book while also choosing the quantity to add.
2. The system will display selected book(s), price(s), and quantities.
3. The customer reviews the order and then confirms it.
4. The system makes sure it has the stock available for purchase and if it does initiates the transaction.
5. The system records the order if available and includes the book, price, quantity, and date/time.
6. The system updates the inventory and displays order ids which would be unique to the purchase.

Alternative Outcome: If an item is not available, the system will tell the customer to adjust the quantity or item. The transaction will be cancelled so the inventory does not change.

Postcondition: Order is saved and inventory is updated. If inventory is not saved, then order is not saved and vice versa.

Use Case 3: Manage inventory and books

Actors: Staff members who have authorization

Precondition: The staff member is logged in with permission to modify inventory and books.

Steps:

1. The staff member searches for the book

2. The staff member updates the quantity of a book or the book of information/description.

3. The staff member can add a book by entering the book title, price, quantity, and ISBN.

4. The system validates the information and saves the changes to the system.

5. The system displays the updated book or quantity.

Use Case 4: Borrow and Return Books (Lending Management)

Actors: Patron (borrowing), Staff Member(borrowing

Precondition: The patron account is active, in good standing, and authenticated in the system.

Steps (Borrowing):

1. The patron looks up an available book in the catalog and selects the borrow option.

2. The system checks that the book copy is currently available and that the patron has not exceeded the maxx limit.

3.The system generates a loan transaction record with the borrower ID, book ID, borrow timestamp, and calculated due date (two weeks).

4. The system marks the selected copy as checked out and updates inventory.

Steps (Returning):

1. The staff member looks up the book by scanning or entering the ISBN/barcode upon return.

2. The system locates the active loan record and updates the status to returned with the current timestamp.

3. The system marks the book copy as available again in inventory.

4. If the return date exceeds the due date, the system records an overdue fee on the patron's account.

Alternative Outcome: If all copies of a book are currently on loan, the system notifies the user that borrowing is unavailable and prevents checkout.

Postcondition: The loan record is stored and circulation availability is accurately updated.

Nonfunctional Requirements

Performance: We aim for our databases' query response time to be relatively low, ideally returning results within 500ms of a user's query. Another performance-based requirement for the bookstore database is that it must be able to handle 50 concurrent read/write transactions without delay. These requirements can be met through a well-designed database structure and will ultimately allow users to have a quick browsing experience.

Security: It is critical that the bookstore database is secure for both the customers and the employee's sake, so we will implement a Role-Based Access Control (RBAC) design. This means that the customers will be limited to having read only access to the store's items and reviews. The only write access the customers would have would pertain to their own orders and reviews. On the other hand, the stores staff would have separate access privileges where they can modify

the stock quantities and scan barcodes. This level of security can be achieved by confirming the user's permissions through their account type when logging in.

Usability: The goal for the bookstore website is to be able to sell as much product as possible, so it is important to maximize the database's usability. When searching a database, it is easy to have certain items overlooked because of a typo or whitespace in a query. To avoid these minor errors that could cost us a sale, we intend on implementing search mechanisms that support case-insensitivity, trailing/leading whitespace, and returning results that contain key words searched. This requirement will prevent empty results from searches that have minor input variations from what we have listed in our database.

Example Usage Scenarios

Example scenario 1: Customer logs-in, they search for a book via title or author, select the desired book, choose a quantity, book delivery type, and then they place the order. The system will then display the price total and order details before confirming the customer's purchase. When completed, the order is stored so the customer and the staff can reference it.

Example Scenario 2: A staff member logs into the program with their employee permissions, scans a book barcode, gets to view the current stock quantity, and update inventory after receiving new copies of books. The updated stock is then available to see it instantly online.

Example Scenario 3: A customer searches for a book, but with incorrect spelling or special characters. The system should still return relevant matching books using the search mechanisms described above in the usability requirements. The customer should still be able to find the book they were looking for and continue on with their browsing.